

Leveraging Visual Signals for Robust Token-Level Uncertainty in Vision–Language Generation

Joseph Hoche

AMIAD, Pôle Recherche, Palaiseau

David Brellmann

Safran Tech

Gianni Franchi

AMIAD, Pôle Recherche, Palaiseau

Abstract

Uncertainty quantification (UQ) remains a critical challenge in Large Vision Language Models (LVLMs) for reliable predictions and real-world deployment. However, most existing methods are adapted from the LLM literature and primarily focus on the language modality, leaving the contribution of visual information to LVLM uncertainty largely underexplored. In this paper, we investigate how LVLMs process visual information and whether this process can be used to improve uncertainty estimation. By analyzing hidden representations after the integration of visual features during the generation process, we observe that high-confidence predictions rely more heavily on visual content than uncertain ones. Building on this insight, we propose **Visual-Grounded Token UQ (VIG-TUQ)**, a training-free framework that explicitly incorporates visual grounding into uncertainty estimation by weighting token-level language uncertainty with visual grounding scores. We evaluate VIG-TUQ on multiple datasets and across diverse LVLM architectures, including early-fusion, late-fusion, and native-fusion models. Results indicate that our method often improves upon existing token-level uncertainty approaches. Code and data will be made available upon acceptance.

1 Introduction

Large Vision-Language Models (LVLMs) extend Large Language Models (LLMs) with visual perception and excel across a wide spectrum of multimodal tasks requiring visual and language understanding (OpenAI, 2024; Gemini, 2024; Guo et al., 2025; OpenAI, 2025; Novikov et al., 2025). Despite their impressive abilities, LVLMs are not yet reliable. In particular, they remain prone to hallucinations, i.e., they may generate incorrect, incomplete, fabricated, or misleading content that appears plausible (Ji et al., 2023; Liu et al., 2024a; Huang et al., 2025). Such behavior limits their use in safety-critical applications like autonomous driving (Gao et al., 2025), robotics (Han et al., 2025) or medicine (Jiang et al., 2024). A promising direction for improving deployment reliability is uncertainty quantification (UQ) to detect when generated content may be unreliable (Hernandez-Lobato & Adams, 2015; Gal & Ghahramani, 2016; Abdar et al., 2021). Prior works show promising results for UQ methods in the language setting for LLMs (Farquhar et al., 2024; Nikitin et al., 2024; Chen et al., 2024). Although these approaches can be extended to LVLMs (Zhang et al., 2024; Lau et al., 2025), their extension is not straightforward. LVLMs operate across multiple modalities and their uncertainty may stem from both linguistic and visual sources. Recent studies have highlighted the tendency of LVLMs to prioritize internal textual knowledge over external visual information (Lee et al., 2025a; Luo et al., 2025; Fu et al., 2025; Long et al., 2026). For instance, given an image depicting a blue orange and the question “*what color is the orange?*”, an LVLM may confidently answer “*orange*”, basing its answer on its parametric knowledge and language prior rather than on the actual visual input. This mismatch motivates the importance of developing specific approaches designed for LVLMs that explicitly capture and quantify the contribution of visual information to LVLM predictions.

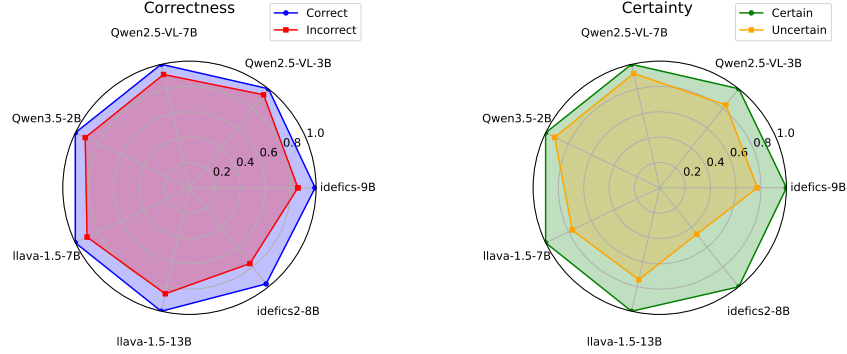


Figure 1: **Correct / confident predictions depend more strongly on visual information than incorrect / uncertain ones.** Radial values indicate the cosine distance between hidden representations from two forward passes: one with visual input and one without. Results are averaged on the **OKVQA** dataset (Marino et al., 2019). The cosine distance measures the model reliance on visual information, where larger values indicate a stronger influence of the image on the internal representation.

In this paper, we investigate how LVLMs process visual information during next-token prediction and whether this process can be leveraged to improve token-level uncertainty estimation. By analyzing hidden representations during generation, we observe that high-confidence predictions rely more heavily on visual content than uncertain ones (Figure 1). Building on this insight, we propose **Visual-Grounding Token Uncertainty Quantification (VIG-TUQ)**, a training-free uncertainty quantification framework for LVLMs. VIG-TUQ explicitly incorporates visual grounding into uncertainty estimation by weighting token-level language uncertainty with visual grounding scores (Figure 2). Specifically, VIG-TUQ quantifies the influence of visual content on next-token prediction using two complementary visual grounding strategies: (1) comparing next-token distributions with and without visual input, and (2) leveraging visual attention weights over image tokens. We hypothesize that tokens more strongly grounded in visual content play a greater role in uncertainty estimation. Our experimental results support this hypothesis and highlight the importance of combining multiple visual grounding signals, as their effectiveness can be model-dependent and capture distinct aspects of visual reasoning. Finally, our approach provides a fine-grained into which specific spans or entities contribute to the uncertainty, while remaining computationally efficient by avoiding reliance on external models or sampling.

Contributions. This work makes the following key contributions: (1) We observe that LVLMs tend to be more confident when their predictions rely more strongly on visual input. (2) We introduce a new training-free uncertainty quantification framework that combines different complementary visual grounding scores to weight token-level uncertainty measures. We evaluate VIG-TUQ on multiple datasets and across diverse LVLm architectures, including early-fusion, late-fusion, and native-fusion models. Results show that our method often improves upon existing token-level uncertainty approaches and remain computationally efficient due to the absence of external models or sampling. (3) We analyze whether uncertainty is uniformly distributed across generated tokens or concentrated within a subset of informative tokens. Our findings indicate that visual information helps identify the tokens most relevant for uncertainty estimation, enabling more effective token selection in LVLMs for uncertainty estimation.

2 Related work

Several recent methods have been proposed to enhance reliability by estimating uncertainty in the predictions of LLMs and LVLMs. These methods are broadly categorized as follows (Kang et al., 2025): (1) logit-based methods, which directly use token probabilities/logits as uncertainty scores (Malinin & Gales, 2021b; Fomicheva et al., 2020); (2) semantic-based methods, which quantifies uncertainty by assessing the semantic consistency of multiple generated answers (Kuhn et al., 2023a; Farquhar et al., 2024; Zhang et al., 2024; Lin et al., 2024; Grewal et al., 2024; Lau et al., 2025; Joo & Cho, 2025); (3) verbalized-based methods, which estimate uncertainty by directly querying the LLM about the reliability of its own generations (Kadavath et al., 2022b; Yona et al., 2024; Xiong et al., 2024); and (4) internal-state-based methods, which analyze internal signals

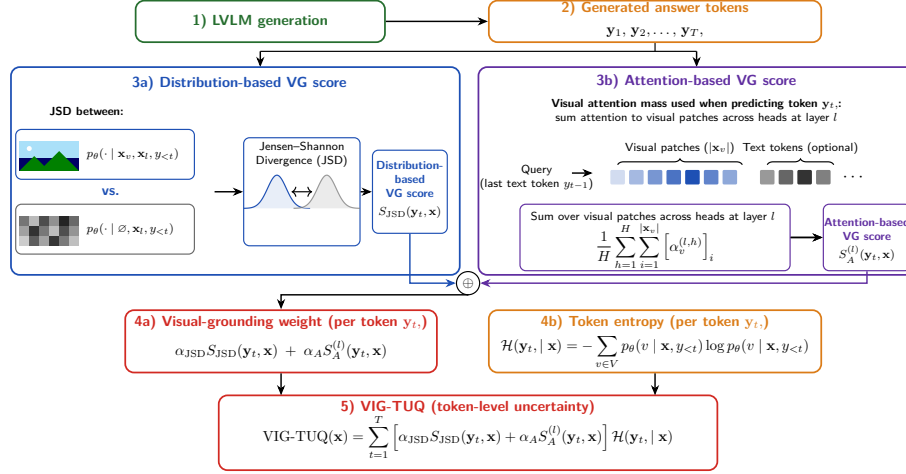


Figure 2: **Overview of the VIG-TUQ pipeline.** VIG-TUQ weights token-level language uncertainty with visual grounding scores using two complementary strategies: a distribution-based score obtained from the Jensen–Shannon divergence between predictions with and without the image, and an attention-based score measuring the attention mass assigned to visual patches.

from LLMs after a single generation to estimate uncertainty, such as features at intermediate layers, attention weights, or other latent representations (Azaria & Mitchell, 2023; Orgad et al., 2025; Kim et al., 2025). Methods listed above may rely on external models (e.g., sentence embedding models (Grewal et al., 2024; Abdaljalil et al., 2025), NLI models (Fadeeva et al., 2024; Nikitin et al., 2024), instruction-tuned LLMs (Farquhar et al., 2024; Ji et al., 2025b)) or involve training dedicated models to estimate uncertainty (Fieback et al., 2024; Orgad et al., 2025; Hoche et al., 2025). In this paper, we propose a training-free hybrid approach to uncertainty estimation that combines a logit-based component to capture linguistic uncertainty with an internal-state-based component to incorporate visual uncertainty. By explicitly leveraging visual information while maintaining computational efficiency with a single-pass and token-level estimation, we introduce a multimodal-aware approach designed for LVLms. Additional technical details and relevant results from prior work are discussed throughout the paper.

3 Preliminaries

Notations. Let $\mathcal{X}_v$ denote the visual input space, $\mathcal{X}_l$ the linguistic input space, and $\mathcal{V}$ the LVLm vocabulary. A multimodal input is depicted as $\mathbf{x} = (\mathbf{x}_v, \mathbf{x}_l)$, where the visual component $\mathbf{x}_v \in \mathcal{X}_v$ consists of a sequence of visual patches and the textual component $\mathbf{x}_l \in \mathcal{X}_l$ consists of a sequence of textual tokens.

An LVLm defines a parameterized function $f_\theta : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$ to encode the visual-linguistic context into a latent representation. In practice, $f_\theta(\cdot, \cdot)$ typically includes a transformer decoder architecture, potentially combined with separate encoders for each modality (see Section 3.1). For a given multimodal input $\mathbf{x} = (\mathbf{x}_v, \mathbf{x}_l)$, the model then autoregressively generates a token sequence $\mathbf{y} = [\mathbf{y}_t]_{t=1}^T$, where each token $\mathbf{y}_t$ is sampled according to:

$$p_\theta(\mathbf{y}_t | \mathbf{x}, \mathbf{y}_{<t}) = \text{softmax}\left(f_h\left(f_\theta(\mathbf{x}_v, [\mathbf{x}_l | \mathbf{y}_{<t}])\right)\right), \quad (1)$$

where $f_h : \mathbb{R}^d \rightarrow \mathbb{R}^{|\mathcal{V}|}$ denotes the output head that projects the latent representation to logits over the vocabulary $\mathcal{V}$ and $\mathbf{y}_{<t} = [\mathbf{y}_1, \dots, \mathbf{y}_{t-1}]$ depicts the previously generated tokens.

3.1 Modality fusion strategies

Given an input $\mathbf{x} = (\mathbf{x}_v, \mathbf{x}_l)$, an LVLM can be described by the approach it uses to combine information from the visual modality $\mathbf{x}_v \in \mathcal{X}_v$ and the textual modality $\mathbf{x}_l \in \mathcal{X}_l$ within the parameterized function $f_\theta : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$.

Native fusion. In native fusion, the modalities are directly combined into a single representation before being processed by the transformer decoder architecture. In particular, the parametric function $f_\theta : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$ is expressed as a composition $f_\theta = f_L \circ f_{L-1} \circ \dots \circ f_1 \circ f_0$, where $f_0 : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$ is the input function and $\{f_i : \mathbb{R}^d \rightarrow \mathbb{R}^d\}_{i=1}^L$ are L decoder transformer layers. The input function $f_0(\cdot, \cdot)$ processes image patches and text tokens independently as:

$$f_0(\mathbf{x}_v, \mathbf{x}_l) = [E_v(\mathbf{x}_v) \parallel E_l(\mathbf{x}_l)], \quad (2)$$

where $E_v(\cdot)$ and $E_l(\cdot)$ map visual patches and language tokens to their embeddings via linear transformations. In these architectures, the model is trained end-to-end from scratch to learn multimodal representations natively within the transformer blocks. LVLM like Qwen models (Qwen Team, 2026) adopts a native fusion strategy.

Early fusion. In early fusion, features from each modality are first extracted separately using dedicated encoders, and then integrated into a shared representation before being processed by the transformer decoder architecture. In particular, the parametric function $f_\theta : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$ is defined as a composition $f_\theta = f_L \circ f_{L-1} \circ \dots \circ f_1 \circ f_0$, where $\{f_i : \mathbb{R}^d \rightarrow \mathbb{R}^d\}_{i=1}^L$ are L decoder transformer layers and $f_0 : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$ is the input function defined as:

$$f_0(\mathbf{x}_v, \mathbf{x}_l) = m_0(E_{\theta_v}(\mathbf{x}_v), E_l(\mathbf{x}_l)); \quad (3)$$

with $m_0(\cdot, \cdot)$ a multimodal fusion module (e.g., concatenation, linear layer, cross-attention layer); $E_{\theta_v}(\cdot)$ a visual encoder with parameters θ_v (e.g., ViT, CNNs); and $E_l(\cdot)$ a text embedding function. Unlike native fusion, the visual encoder $E_{\theta_v}(\cdot)$ is a pre-trained model that can be optimized independently. LVLMs like LLaVA (Liu et al., 2023) and Qwen-VL (Bai et al., 2024) adopt an early fusion approach. Further details on early fusion, fusion modules, and unimodal encoders can be found in (Li & Tang, 2024; An et al., 2025).

Late fusion. Late fusion also starts with the extraction of unimodal features using dedicated encoders. However, unlike early fusion, the integration between modalities occurs within the intermediate representations of the transformer decoder architecture rather than before them. In particular, the parametric function $f_\theta : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$ can be defined recursively over L layers as

$$\begin{aligned} f_\theta(\mathbf{x}_v, \mathbf{x}_l) &= f_L(\mathbf{h}_{L-1}, \mathbf{x}_v), \quad \text{where} \\ \mathbf{h}_i &= f_i(\mathbf{h}_{i-1}, \mathbf{x}_v) = g_i(m_i(E_{\theta_v^{(i)}}(\mathbf{x}_v), \mathbf{h}_{i-1})) \quad (1 \leq i \leq L) \\ \mathbf{h}_0 &= E_l(\mathbf{x}_l); \end{aligned} \quad (4)$$

with $E_l(\cdot)$ a text embedding function. For each layer i , $g_i : \mathbb{R}^d \rightarrow \mathbb{R}^d$ depicts a transformer decoder layer, $m_i(\cdot, \cdot)$ denotes a multimodal fusion module (e.g., concatenation, linear layer, cross-attention layer), and $E_{\theta_v^{(i)}}(\cdot)$ is a visual encoder with parameters $\theta_v^{(i)}$ (e.g., ViT, CNNs). Note that the vision modality is not necessarily integrated at every layer and the fusion module can simply preserve the hidden representation, i.e., $m_i(E_{\theta_v^{(i)}}(\mathbf{x}_v), \mathbf{h}_i) = \mathbf{h}_i$. Similarly to early fusion, visual encoders $E_{\theta_v^{(i)}}(\cdot)$ are pre-trained models that can be optimized independently. LVLMs like IDEFICS (Laurençon et al., 2023) and Flamingo (Alayrac et al., 2022) adopt such approach. Further details on late fusion, fusion modules, and unimodal encoders can be found in (Li & Tang, 2024; An et al., 2025).

3.2 Token-level UQ methods

Given a multimodal input $\mathbf{x}$, token-level UQ methods assign an uncertainty score s_t to each generated token $\mathbf{y}_t$ (equation 1), with a higher score indicating greater uncertainty. The uncertainty score $s(\cdot)$ for a generated sequence $\mathbf{y} = [\mathbf{y}_t]_{t=1}^T$ can be derived by aggregating the token-level scores of each token $\mathbf{y}_t$, e.g., by averaging across tokens $s(\mathbf{y}) = \frac{1}{T} \sum_{t=1}^T s_t$ or by taking their maximum $s(\mathbf{y}) = \max_t s_t$.

Negative Log-Likelihood (Log-Perplexity). A straightforward token-level uncertainty score is the token negative log-likelihood (Fomicheva et al., 2020; Malinin & Gales, 2021a):

$$\text{NLL}(\mathbf{y}_t, \mathbf{x}, \mathbf{y}_{<t}) = -\log p_{\theta}(\mathbf{y}_t \mid \mathbf{x}, \mathbf{y}_{<t}). \quad (5)$$

Token Entropy. Another common token-level score is the token entropy of the autoregressive distribution (Fomicheva et al., 2020; Malinin & Gales, 2021a):

$$\mathcal{H}(\mathbf{y} \mid \mathbf{x}, \mathbf{y}_{<t}) = -\sum_{\mathbf{y} \in \mathcal{V}} p_{\theta}(\mathbf{y} \mid \mathbf{x}, \mathbf{y}_{<t}) \log p_{\theta}(\mathbf{y} \mid \mathbf{x}, \mathbf{y}_{<t}). \quad (6)$$

Claim-Conditioned Probability (CCP). CCP (Fadeeva et al., 2024) estimates token-level uncertainty by measuring how much the top K token alternatives $\{\mathbf{y}_t^k\}_{k=1}^K$ contradict the generated token $\mathbf{y}_t$ using a Natural Language Inference (NLI) model:

$$\text{CCP}(\mathbf{y}_t, \mathbf{x}, \mathbf{y}_{<t}) = -\log \frac{\sum_{k: \text{NLI}(\mathbf{y}_{<t}, \mathbf{y}_t^k, \mathbf{y}_t) = \text{entail}} p_{\theta}(\mathbf{y}_t^k \mid \mathbf{x}, \mathbf{y}_{<t})}{\sum_{k: \text{NLI}(\mathbf{y}_{<t}, \mathbf{y}_t^k, \mathbf{y}_t) \in \{\text{entail}, \text{contradict}\}} p_{\theta}(\mathbf{y}_t^k \mid \mathbf{x}, \mathbf{y}_{<t})}, \quad (7)$$

where $\text{NLI}(\cdot)$ determines whether concatenating $\mathbf{y}_t^k$ with the preceding generated context $\mathbf{y}_{<t}$ entails (semantic equivalence) or contradicts (different meaning) the generated content $\mathbf{y}_{<t} \cup \mathbf{y}_t$.

Token-Shifting Attention to more Relevant (Token-SAR). Token-SAR (Duan et al., 2024) defines a token-level uncertainty score by reweighting token likelihood with a semantic relevance factor as:

$$\text{Token-SAR}(\mathbf{y}_t, \mathbf{x}, \mathbf{y}_{<t}) = -\log p_{\theta}(\mathbf{y}_t \mid \mathbf{x}, \mathbf{y}_{<t}) \tilde{R}(\mathbf{x}, \mathbf{y}, \mathbf{y}_t), \quad (8)$$

where $\tilde{R}(\mathbf{x}, \mathbf{y}, \mathbf{y}_t) = R(\mathbf{x}, \mathbf{y}, \mathbf{y}_t) / \sum_{i=1}^T R(\mathbf{x}, \mathbf{y}, \mathbf{y}_i)$ is the normalized relevance score with

$$R(\mathbf{x}, \mathbf{y}, \mathbf{y}_t) = 1 - |g(\mathbf{x} \cup \mathbf{y}, \mathbf{x} \cup \mathbf{y} \setminus \{\mathbf{y}_t\})|$$

the relevance that quantifies how important $\mathbf{y}_t$ is in reflecting the semantics of $\mathbf{y}$ by comparing the semantic change before and after removing this token and $g(\cdot, \cdot)$ a similarity between two inputs on a scale of -1 to 1 .

4 Method

In this section, we introduce Visual-Grounding Token Uncertainty Quantification (VIG-TUQ), a training-free UQ framework for LVLMS that weights token-level language uncertainty with visual grounding scores. We first discuss our motivations (Section 4.1), then describe the visual grounding score used (Section 4.2), and finally introduce the proposed method (Section 4.3).

4.1 Motivations

A recent line of works highlights the tendency of LVLMS to prioritize internal textual knowledge over external visual information (Lee et al., 2025a; Luo et al., 2025; Fu et al., 2025; Long et al., 2026). This suggests that visual grounding could play a crucial role in identifying unreliable responses. To investigate this phenomenon and better understand the contribution of visual information to token generations, we conduct a pilot study on how LVLMS process visual information. Specifically, we compare hidden representations (via the cosine distance) obtained from two independent forward passes: one with visual evidence and without it. This analysis is performed on generated sentences from the **OKVQA** dataset (Marino et al., 2019) across different LVLM architectures (Long et al., 2026) using greedy decoding with visual evidence. (further details are provided in Appendix F). Figure 1 summarizes our results and highlights that correct and confident predictions rely more on the visual content than incorrect and uncertain ones. Furthermore, experiments reported in Appendix C and Appendix F indicate that the extent to which visual information is processed depends on both the training setup and the model architecture (see Section 3). These observations motivate the introduction of multiple visual grounding scores to capture different aspects of visual understanding. Overall, our results suggest that tokens whose generation strongly depends on visual input are more informative and should therefore be assigned greater weight in uncertainty estimation. Building on this insight, we introduce different visual grounding token-level strategies and our VIG-TUQ approach in the following sections.

4.2 Visual grounding scores

We propose two scoring strategies to measure how much an LVLM relies on its visual input when generating a token.

Distribution-based scores. Given a multimodal input $\mathbf{x} = (\mathbf{x}_v, \mathbf{x}_l)$, with visual content $\mathbf{x}_v$ and textual content $\mathbf{x}_l$, this approach quantifies the contribution of the visual input $\mathbf{x}_v$ by measuring how much the next-token distribution changes when the visual information is removed. In particular, for each generated token $\mathbf{y}_t$, we compare the original LVLM next-token distribution $p_\theta(\cdot \mid \mathbf{x}_v, \mathbf{x}_l, \mathbf{y}_{<t})$ with the next-token distribution $p_\theta(\cdot \mid \emptyset, \mathbf{x}_l, \mathbf{y}_{<t})$ when the visual information $\mathbf{x}_v$ is removed using the Jensen-Shannon divergence:

$$d_t(\mathbf{y}, \mathbf{x}) = \text{JSD}(p_\theta(\cdot \mid \mathbf{x}_v, \mathbf{x}_l, \mathbf{y}_{<t}) \parallel p_\theta(\cdot \mid \emptyset, \mathbf{x}_l, \mathbf{y}_{<t})). \quad (9)$$

A larger value of $d_t(\mathbf{y}, \mathbf{x})$ indicates that removing the visual input induces a stronger distributional shift when generating $\mathbf{y}_t$ and thus a greater reliance on visual information. To obtain a relative token-level importance score, we normalize the divergence values across all generated tokens: $S_{\text{JSD}}(\mathbf{y}_t, \mathbf{x})$, where

$$S_{\text{JSD}}(\mathbf{y}_t, \mathbf{x}) = \frac{d_t(\mathbf{y}_t, \mathbf{x})}{\sum_{i=1}^T d_t(\mathbf{y}_i, \mathbf{x})}. \quad (10)$$

This normalized score captures the relative contribution of the visual input to each generated token $\mathbf{y}_t$ with respect to the full sequence.

Attention-based Scores. The attention mechanism provides another way to estimate how much the visual content contributes to the next-token prediction. For instance, given a multimodal input $\mathbf{x} = (\mathbf{x}_v, \mathbf{x}_l)$ with a visual content $\mathbf{x}_v$ and a textual content $\mathbf{x}_l$, we could use the attention weight vector used for predicting the next token in the l -th layer and the h -th attention head that interacts with the visual content $\mathbf{x}_v$:

$$\boldsymbol{\alpha}_t^{(l,h)} = \text{softmax}\left(\frac{\mathbf{K}_t^{(l,h)} \mathbf{q}_t^{(l,h)}}{\sqrt{d_k}}\right) \in [0, 1]^N, \quad (11)$$

where $\mathbf{q}^{(l,h)} \in \mathbb{R}^{d_k}$ is the query vector corresponding to the last textual token in $\mathbf{x}_l$; $\mathbf{K}^{(l,h)} \in \mathbb{R}^{N \times d_k}$ is the key matrix featuring the visual content $\mathbf{x}_v$ and possibly the textual content $\mathbf{x}_l$ (in the case of self-attention); d_k is the dimensionality of the key and query vectors; and N is the number of visual (and possibly textual) elements in $\mathbf{x}$. We denote by $\boldsymbol{\alpha}_{t,v}^{(l,h)}$ the subset of attention weights interacting with visual tokens or patches. For cross-attention heads, $\boldsymbol{\alpha}_{t,v}^{(l,h)} = \boldsymbol{\alpha}_t^{(l,h)}$, while for self-attention heads, $\boldsymbol{\alpha}_{t,v}^{(l,h)} = [\alpha_{t,1}^{(l,h)}, \dots, \alpha_{t,|\mathbf{x}_v|}^{(l,h)}]$, where $|\mathbf{x}_v|$ denotes the number of visual tokens or patches. We then define the attention-based visual grounding score for token $\mathbf{y}_t$ at layer l as

$$a_t^{(l)}(\mathbf{y}, \mathbf{x}) = \frac{1}{H} \sum_{h=1}^H \sum_{i=1}^{|\mathbf{x}_v|} [\boldsymbol{\alpha}_{t,v}^{(l,h)}]_i, \quad (12)$$

where H is the number of attention heads that interact with the visual content. A higher value of $a_t^{(l)}(\mathbf{y}, \mathbf{x})$ indicates stronger dependency on the visual modality, whereas a lower value suggests that visual information contributes minimally to the next-token prediction at that layer. As for the distribution-based score, we normalize these raw attention scores across all generated tokens to obtain a relative token-level importance score $S_A^{(l)}(\mathbf{y}_t, \mathbf{x})$, where

$$S_A(\mathbf{y}_t, \mathbf{x}) = \frac{a_t(\mathbf{y}_t, \mathbf{x})}{\sum_{i=1}^T a_t(\mathbf{y}_i, \mathbf{x})}. \quad (13)$$

This normalized score captures the relative contribution of token $\mathbf{y}_t$ to the overall visual attention mass in the generated answer.

4.3 VIG-TUQ

Our approach leverages visual grounding scores defined in Section 4.2 and combines them with token-level uncertainty scores. Our approach assumes that tokens more strongly grounded in visual content contribute more significantly to uncertainty estimation, as they are more likely to support factual claims in the generated output (see Section 5).

Visual-grounding token UQ. We define VIG-TUQ as a UQ framework for LVLMs that weights token-level uncertainty scores with visual grounding scores. Specifically, given a multimodal input x and a generated answer $y = [y_t]_{t=1}^T$, we define a visual-grounded uncertainty score by combining the distribution-based visual grounding score $S_{\text{JSD}}(\cdot, \cdot)$ (equation 10) and the attention-based visual grounding score $S_A^{(l)}(\cdot, \cdot)$ (equation 13) with the token entropy $\mathcal{H}(y_t | x)$ (equation 6), and summing over all generated tokens:

$$\text{VIG-TUQ}(x) = \sum_{t=1}^T \left[\alpha_{\text{JSD}} S_{\text{JSD}}(y_t, x) + \alpha_A S_A^{(l)}(y_t, x) \right] \mathcal{H}(y_t | x), \quad (14)$$

where $\alpha_{\text{JSD}} > 0$ and $\alpha_A > 0$ are weighting hyperparameters, and l denotes the layer used to compute the attention-based visual grounding score $S_A^{(l)}(\cdot, \cdot)$. This formulation can be interpreted as a measure of predictive uncertainty that explicitly penalizes confidence not grounded in core visual evidence.

5 Experiments

5.1 Experimental setup

Datasets and models. We evaluate our method across a range of visual question answering (VQA) tasks, datasets, and model architectures. Specifically, we conduct experiments on four benchmark datasets: **ADVQA** (Li et al., 2021), **VQARAD** (Lau et al., 2018), **OKVQA** (Marino et al., 2019), and **VizWiz** (Gurari et al., 2018). Our evaluation includes several recent LVLM architectures, namely Qwen2.5-VL-3B and Qwen2.5-VL-7B (Bai et al., 2025), Qwen3.5-2B (Qwen Team, 2026), llava1.5-7B, llava1.5-13B (Liu et al., 2024b), idfics-9B (Laurençon et al., 2023) and idfics2-8B (Laurençon et al., 2023). Additional details regarding the datasets, models, and experimental setup are provided in Appendix A.

Evaluation metrics. Following prior work (Kuhn et al., 2023b; Farquhar et al., 2024; Janiak et al., 2025a; Lau et al., 2025), we evaluate uncertainty estimation methods using the AUROC metric (Hendrycks & Gimpel, 2017) and ECE (Guo et al., 2017). AUROC (Area Under the Receiver Operating Curve) measures the ability of an uncertainty score to distinguish between correct and incorrect model predictions. Higher values indicate better discrimination, with a perfect uncertainty estimator achieving a score of 1, while a random uncertainty measure yields an expected score of 0.5. ECE (Expected Calibration Error) quantifies how well predicted probabilities align with empirical accuracies, where lower values indicate better calibration. Additional details on the evaluation metrics are provided in Appendix A.

Ground truth label. To evaluate the correctness of generated answers, we adopt an *LLM-as-a-Judge* approach, following the recommendation of (Janiak et al., 2025a). For each generated answer, the dataset provides a set of valid ground-truth responses. We use Llama-3.1-8B to determine whether the generated answer is semantically consistent with any of the provided ground-truth answers. This setup has been shown to be reliable and to exhibit strong agreement with human judgments (Janiak et al., 2025b).

Baselines. We compare our method against several widely used token-level (TL) uncertainty baselines as well as two sample-diversity approaches, that are semantic-level (SL). For TL methods, we include Token Entropy (equation 6), Log-Perplexity (equation 5), Max-Prob (Fomicheva et al., 2020; Malinin & Gales, 2021a), CCP 7, and Token-SAR (equation 8). We also include results of SL methods like the Semantic Entropy (SE) (Kuhn et al., 2023b) and the Kernel Language Entropy (KLE) with a Matern kernel (Nikitin et al., 2024).

5.2 Results

Overall performance comparison. Tables 1 and 2 report the performance of the evaluated uncertainty estimation methods. Table 1 presents AUROC scores aggregated across LVLM architectures for each dataset, while Table 2 reports results aggregated across datasets for each model architecture. Across all datasets and models, VIG-TUQ consistently achieves higher AUROC scores than

Table 1: **VIG-TUQ improves AUROC compared to all token-level (TL) baselines and remains competitive with semantic-level (SL) baselines.** Evaluation metrics for datasets averaged across all LVLM architectures. **Bold** values indicates best performance, underline values indicate the second-best among TL.

		OKVQA		VIZWIZ		VQARAD		ADVQA	
		AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$
TL	<i>Log-Perplexity</i>	0.625	0.117	0.645	0.096	0.634	0.185	0.616	0.166
	<i>Token-Sar</i>	0.592	0.139	0.589	0.127	0.614	0.181	0.568	0.171
	<i>CCP</i>	0.628	0.119	0.640	0.104	0.617	0.157	<u>0.631</u>	0.127
	<i>Max-Prob</i>	0.623	0.126	0.640	0.119	0.619	0.194	<u>0.623</u>	0.182
	<i>Token-Entropy</i>	0.629	0.114	0.648	0.094	0.637	0.183	0.613	0.169
	<i>VIG-TUQ-A (ours)</i>	0.649	0.104	0.675	0.086	0.655	0.181	0.631	0.159
	<i>VIG-TUQ-JSD (ours)</i>	0.633	0.121	0.663	0.092	0.639	0.175	0.591	0.166
	<i>VIG-TUQ (ours)</i>	0.655	0.104	0.678	<u>0.089</u>	0.681	<u>0.173</u>	0.632	<u>0.157</u>
SL	<i>SE</i>	0.669	0.078	0.648	0.061	0.644	0.168	0.636	0.194
	<i>KLE</i>	0.691	0.077	0.681	0.064	0.652	0.168	0.648	0.193

VIG-TUQ is as described in Eq14. *VIG-TUQ-A* and *VIG-TUQ-JSD* are variants using only one of the visual grounding criteria.

Table 2: **VIG-TUQ improves AUROC compared to all token-level (TL) baselines and remains competitive with semantic-level (SL) baselines.** Evaluation metrics for LVLM architectures averaged across all vision datasets. **Bold** values indicates best performance. underline values indicate the second-best among TL.

	Method	Qwen2.5-VL-3B		Qwen2.5-VL-7B		llava-1.5-13B		llava-1.5-7B		idefics2-8B		idefics-9B		Qwen3.5-2B	
		AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$	AUROC $\uparrow$	ECE $\downarrow$
TL	<i>Log-Perplexity</i>	0.653	0.128	0.622	0.164	0.646	0.139	0.630	0.135	0.624	0.148	0.625	0.157	0.615	0.111
	<i>Token-Sar</i>	0.554	0.175	0.588	0.183	0.639	0.135	0.609	0.136	0.599	0.153	0.550	0.170	0.587	0.130
	<i>CCP</i>	0.644	0.120	0.620	0.148	0.627	0.137	0.624	0.120	0.627	0.137	0.659	0.118	0.604	0.099
	<i>Max-Prob</i>	0.643	0.150	0.616	0.172	0.625	0.163	0.623	0.159	0.617	0.162	<u>0.656</u>	0.164	0.611	0.112
	<i>Token-Entropy</i>	0.651	0.126	0.622	0.164	0.650	0.140	0.632	0.133	0.627	0.146	0.629	0.154	0.616	0.109
	<i>VIG-TUQ-A (ours)</i>	0.689	0.113	0.651	0.152	0.660	0.134	0.640	0.126	0.645	0.145	0.660	0.142	0.633	0.103
	<i>VIG-TUQ-JSD (ours)</i>	0.652	0.124	0.611	0.160	0.656	0.135	0.652	0.126	0.647	0.145	0.575	0.167	0.614	0.108
	<i>VIG-TUQ (ours)</i>	0.691	0.111	0.654	<u>0.152</u>	0.671	0.133	0.663	<u>0.124</u>	0.661	0.140	0.660	<u>0.142</u>	<u>0.632</u>	<u>0.104</u>
SL	<i>SE</i>	0.662	0.114	0.651	0.148	0.668	0.127	0.674	0.121	0.655	0.125	0.674	0.121	0.555	0.123
	<i>KLE</i>	0.673	0.113	0.658	0.147	0.683	0.127	0.686	0.121	0.668	0.124	0.682	0.120	0.627	0.127

VIG-TUQ is as described in Eq14, *VIG-TUQ-A* and *VIG-TUQ-JSD* are variants using only one of the visual grounding criteria.

existing token-level baselines, demonstrating improved reliability while preserving the computational efficiency of single-pass uncertainty estimation. Although sample-diversity approaches such as SE and KLE still obtain slightly higher AUROC values in some cases, these methods require multiple generations and additional computations. Consequently, they incur a significantly higher inference cost compared to token-level approaches. Our results therefore highlight that VIG-TUQ provides a favorable trade-off between uncertainty estimation quality and computational efficiency.

Visual grounding & informative tokens. Our approach is based on the assumption that visual grounding scores help identify the tokens most relevant for uncertainty estimation. We further hypothesize that the effectiveness of these scores is model-dependent and that they capture distinct aspects of visual reasoning. To test this hypothesis, we compute the sum of token entropy over the top $k\%$ of generated tokens. These tokens are selected according to either the attention-based score the the distribution-based score (equation 10), attention-based score (equation 13), or randomly from generated tokens. Additional details are provided in Appendix B. Results presented in Figure A.4 lead to three main observations. (1) Both visual grounding scores are effective token selectors, as they mostly outperform random selection across models and datasets; (2) Selecting only the most visually grounded tokens is generally more effective than considering all tokens; (3) The attention-based and distribution-based criteria perform better on different models and datasets, which confirm their complementary nature and motivate their combination in VIG-TUQ. Overall, these findings suggest that uncertainty is not uniformly distributed across the generated sequence and that only a subset of tokens carries the most informative signals for uncertainty quantification.

Computational cost. We evaluate the efficiency of VIG-TUQ by comparing its average inference time with existing token-level UQ methods on OKVQA. Table 3 reports the average runtime for answer generation and uncertainty estimation under identical backbone models and hardware settings. Since all token-level methods share the same decoding procedure, generation time remains constant and any runtime differences arise only from the post-hoc UQ computation. Unlike CCP and Token-SAR, which depend on external models and thus introduce additional computational overhead,

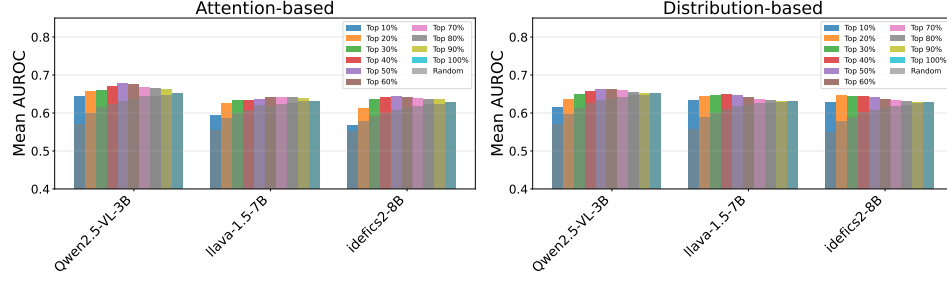


Figure 3: **Visual grounding scores help identify the tokens most relevant for uncertainty estimation.** AUROC performance reported for the sum of top $k\%$ token entropies across different LVLM architectures. Top $k\%$ token entropies are selected according to either the attention-based score or the distribution-based score (equation 10), attention-based score (equation 13), or randomly from generated tokens (details in Appendix B). Gray bars correspond to a random token-selection baseline and averaged over 10 runs. All results are averaged across all datasets.

Table 3: **VIG-TUQ is computationally efficient.** This table shows the average inference time for: answer generation and uncertainty quantification on the *OKVQA* dataset.

Model	Log-Perplexity	Max-Prob	Token-Entropy	CCP	Token-SAR	VIG-TUQ	KLE
Qwen2.5-3B	1.04s	0.99s	1.02s	3.31s	1.65s	1.32s	4.70s
llava-1.5-7B	0.85s	0.83s	0.81s	2.75s	1.13s	0.98s	4.90s

VIG-TUQ is more efficient because it leverages only token probabilities and attention weights that are computed by the generation model itself. Although lightweight baselines such as Token-Entropy, Log-Perplexity, and Max-Prob remain slightly faster, VIG-TUQ incurs only minimal overhead while being substantially more efficient than other token-level UQ approaches.

6 Conclusion

In this paper, we present a token-level analysis of how reliance on visual evidence in LVLMs affects uncertainty quantification. Our findings indicate that visual information helps identify the most relevant token for uncertainty quantification and underscore the importance of combining multiple visual grounding signals, as their effectiveness can vary across models and capture complementary aspects of visual reasoning. Building on these insights, we introduce Visual-Grounded Token UQ (VIG-TUQ), a training-free and computationally efficient framework that explicitly incorporates visual grounding into uncertainty estimation by weighting token-level language uncertainty with visual grounding scores. Experiments on multiple datasets and across diverse LVLM architectures (including early-fusion, late-fusion, and native-fusion models) indicate that our method often improves upon existing token-level uncertainty approaches. We hope our work will encourage future research on vision-aware and fine-grained uncertainty quantification for LVLMs.

Limitations. While our approach improves upon existing token-level uncertainty methods, semantic uncertainty techniques still achieve slightly higher AUROC scores in certain cases. Future work could explore hybrid approaches that integrate semantic uncertainty with visual grounding signals to leverage the strengths of both approaches. Additionally, our method is currently limited to white-box models as it requires access to internal representations.

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Leveraging Visual Signals for Robust Token-Level Uncertainty in Vision–Language Generation

–Supplementary Material–

A Experimental protocol

A.1 Datasets

We evaluate VIG-TUQ on several Visual Question Answering (VQA) datasets: **ADVQA** (Li et al., 2021), which consists of a training set of 5,000 samples and a test set of 2,000 samples, both drawn from the original training set; **VQARAD** (Lau et al., 2018), which includes 2,000 samples from the original training set and 500 samples from the original test set; **OKVQA** (Marino et al., 2019), which contains 5,000 training samples drawn from the original training set and 2,000 test samples from the validation set; and **VizWiz** (Gurari et al., 2018), which contains 5,000 training samples and 2,000 test samples drawn from the original training set.

A.2 Models

We evaluate VIG-TUQ on widely used Large Vision Language Models (LVLMs). In particular, we conduct experiments on Qwen2.5-VL-3B and Qwen2.5-VL-7B (Bai et al., 2025), Qwen3.5-2B (Qwen Team, 2026), LLaVa1.5-7B, LLaVa1.5-13B (Liu et al., 2024b), idefics-9B (Laurençon et al., 2023) and idefics2-8B (Laurençon et al., 2023). We include models with different architectures to cover the range of modality fusion strategies discussed in Section 3.1 and to analyze how these fusion strategies affect the visual grounding. In particular, llava, idefics2 and Qwen-VL models adopt a *early fusion* approach, while idefics-9B model rely on an *late fusion* approach and Qwen3.5 on a *native* fusion one.

A.3 Metrics

AUROC. We assign a probability score to each example for belonging to the positive class (positive reflects that the LVLM is certain about its answer). Area Under the ROC Curve (AUROC) (Hendrycks & Gimpel, 2017) measures how well these scores rank true positives above true negatives. It is the chance that a randomly chosen positive example receives a higher score than a randomly chosen negative one. Higher AUROC means better class separation.

ECE. Expected Calibration Error (ECE) (Guo et al., 2017) measures how well a model’s predicted confidence aligns with its empirical accuracy. A low ECE indicates that the model is well calibrated, meaning that predictions made with confidence c are correct approximately c fraction of the time. Conversely, a high ECE indicates miscalibration, for example when the model is systematically overconfident or underconfident. Since ECE is defined for confidence scores in $[0, 1]$, we first normalize all UQ scores to this range before computing ECE.

B Token selection analysis

In Section 5, to evaluate whether visual grounding scores help identify the tokens most relevant for uncertainty estimation, we conduct a token selection experiment. For each generated answer $\mathbf{y} = [\mathbf{y}_1, \dots, \mathbf{y}_T]$, we rank token entropies $\mathcal{H}(\mathbf{y}_t | \mathbf{x})$ (equation 6) according to either the distribution-based grounding score $S_{\text{JSD}}(\mathbf{y}_t, \mathbf{x})$ (equation 10) or the attention-based grounding score $S_A^{(l)}(\mathbf{y}_t, \mathbf{x})$ (equation 13). We then retain only the top $k\%$ of token entropies according to each criterion and compute an overall uncertainty score by summing the selected token entropy values. We also repeat the same procedure using a random subset of $k\%$ generated tokens instead of visually grounded tokens. To reduce the bias of this random selection, random token selection is repeated 10 times for each model, dataset, and value of k and we report the AUROC average. Figure A.4 depicts results across different LVLM architectures.

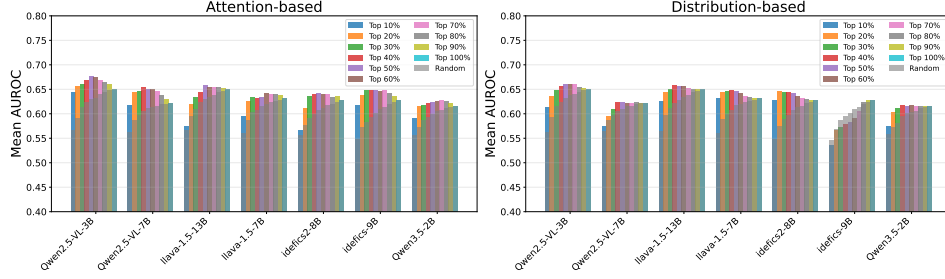


Figure A.4: **Visual grounding scores help identify the tokens most relevant for uncertainty estimation.** AUROC performance reported for the sum of top $k\%$ token entropies across different LVLM architectures. Top $k\%$ token entropies are selected according to either the attention-based score or the distribution-based score (equation 10), attention-based score (equation 13), or randomly from generated tokens. Gray bars correspond to a random token-selection baseline and averaged over 10 runs. All results are averaged across all datasets.

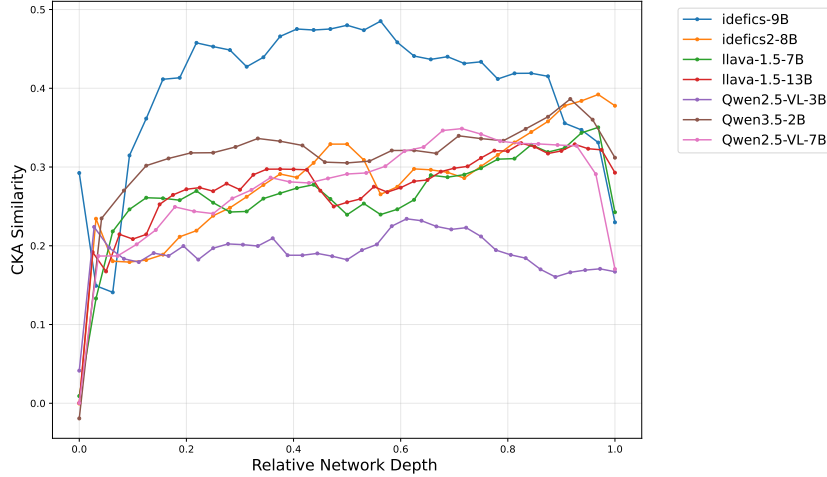


Figure A.5: CKA similarity between LVLM hidden states and reference vision features across normalized network depth. For each model, we compare layer-wise hidden representations to the final-layer representation of a frozen vision encoder. Higher CKA values indicate stronger alignment with the original visual representation. The late-fusion model `idefics-9B` exhibits the highest visual alignment across much of the network, consistent with architectures that continue to inject visual information throughout the transformer layers.

C Additional experiments

Impact of LVLM architecture. Figure A.5 aims to analyze how different LVLM architectures use visual information across their layers. To do so, we compute the Centered Kernel Alignment (CKA) similarity between the hidden representations of each LVLM (on **OKVQA**) and the final-layer features of a reference vision encoder, DINOv2 (Oquab et al., 2023). We use DINOv2 as a reference because it is trained for visual representation learning and captures strong image features across many vision tasks. As a result, it provides a useful proxy for purely visual information. We therefore measure how similar LVLM representations are to these visual features. For each LVLM, we extract hidden states across all transformer layers and compare them to the reference visual features. Because the evaluated models have different numbers of layers, we normalize layer indices to a relative depth scale from 0.0, corresponding to the embedding layer, to 1.0, corresponding to the final transformer layer. This enables a direct comparison of where visual information is most strongly reflected across architectures.

The resulting plot shows how closely each model’s internal representations align with the original visual features as generation progresses through the network. High CKA similarity indicates that the model representation remains strongly related to the visual encoder representation, whereas lower similarity suggests a shift toward language- or task-specific representations. We observe that `idefics-9B`, the only late-fusion architecture in this comparison, exhibits the strongest similarity to the reference vision encoder. This behavior is consistent with its architectural design: unlike early-fusion models, which inject visual features mainly before the language-model layers, late-fusion models continue to integrate visual components throughout the network, leading to stronger visual alignment at intermediate or deeper layers.

D Complete results

In this section, we present the detailed performance of all UQ methods across datasets and models described in Appendix A. Table A.4 reports the AUROC scores achieved by each method on every dataset and LVLm architecture.

Comparison with token-level UQ methods. VIG-TUQ consistently outperforms conventional token-level uncertainty metrics, including Log-Perplexity, Max-Prob, Token-Entropy, CCP, and Token-SAR. For instance, on the **OKVQA** dataset, VIG-TUQ achieves the highest AUROC score. Similar trends are observed for **VQARAD** and **ADVQA**, where VIG-TUQ ranks among the top-performing methods. These results confirm that integrating multiple importance scores and visual grounding scores improves the performance of token-level UQ methods.

Comparison with consistency-based methods. Semantic Entropy (SE) (Farquhar et al., 2024) and Kernel Language Entropy (KLE) (Nikitin et al., 2024) achieve slightly higher AUROC on certain datasets. This behavior is expected as these methods estimate uncertainty by measuring the semantic consistency of generated responses, which provides a more comprehensive assessment of uncertainty across the entire generated content. However, these methods require multiple sampling, which also makes them significantly more expensive at inference time (see Table 3). In contrast, our approach requires only a single or two samplings and therefore remains much more efficient while still maintaining competitive performance.

Effect across datasets. We also observe that the performance of UQ methods varies across datasets. For instance, VIG-TUQ clearly outperforms on **OKVQA** and **VQARAD**, which require stronger reasoning and better exploitation of the visual content, which makes reliable uncertainty estimation more challenging. On **VizWiz**, the improvements are still visible but slightly smaller. This dataset contains real-world images with visual noise, which can make the visual grounding more difficult for all models.

Effect across model architectures. The benefits of VIG-TUQ are consistent across different LVLm architectures. This suggests that VIG-TUQ is not tied to a specific architecture and can be effectively applied across a wide range of multimodal architectures.

E Hyperparameters

The hyperparameters used in VIG-TUQ are reported in Table A.6. They are selected on a small training split for each dataset–model combination. To limit the search space and avoid overfitting to the validation setting, we impose a strict constraint on the weighting coefficients: both α_{JSD} and α_A are restricted to integer values in $[0, 5]$. This deliberately coarse search is intended to ensure that the performance gains of VIG-TUQ do not rely on extensive hyperparameter tuning. The selected hyperparameters show that some model- and dataset-specific adaptation is beneficial. This is consistent with the token selection analysis in Figure A.4, where the attention-based and distribution-based visual grounding scores do not perform best in the same settings. For example, for `llava-1.5-7B` on **VQARAD**, VIG-TUQ relies only on the distribution-based score, whereas for `Qwen3.5-2B` on **OKVQA**, it relies only on the attention-based score. Nevertheless, in most model–dataset combinations, both scores receive nonzero weights, suggesting that the two grounding criteria provide complementary information for uncertainty quantification.

Table A.4: **VIG-TUQ outperforms token-level UQ baselines.** Evaluation metrics for LVLMM architectures across all datasets and models.

	Qwen2.5-VL-3B [ECL]	Qwen2.5-VL-3B [ECL]	Qwen2.5-VL-7B [AIROCT]	Qwen2.5-VL-7B [ECL]	llava-1.5-13B [AIROCT]	llava-1.5-13B [ECL]	llava-1.5-7B [AIROCT]	llava-1.5-7B [ECL]	idefics2-8B [AIROCT]	idefics2-8B [ECL]	idefics-9B [AIROCT]	idefics-9B [ECL]	Qwen3.5-2B [AIROCT]	Qwen3.5-2B [ECL]
OKVQA	<i>Log-Perplexity</i>	0.610	0.110	0.594	0.108	0.635	0.137	0.634	0.116	0.637	0.146	0.605	0.104	0.649
	<i>Token-Sar</i>	0.551	0.158	0.580	0.115	0.639	0.145	0.637	0.112	0.603	0.159	0.524	0.157	0.600
	<i>CCP</i>	0.607	0.111	0.611	0.087	0.614	0.167	0.612	0.136	0.647	0.170	0.642	0.065	0.625
	<i>Max-Pmb</i>	0.592	0.151	0.608	0.111	0.611	0.147	0.609	0.152	0.634	0.138	0.644	0.105	0.649
	<i>Token-Entropy</i>	0.609	0.108	0.601	0.100	0.641	0.136	0.638	0.114	0.642	0.145	0.608	0.096	0.651
	<i>VIG-TUQ-A (ours)</i>	0.657	0.076	0.620	0.091	0.644	0.135	0.651	0.104	0.664	0.143	0.645	0.070	0.653
	<i>VIG-TUQ-JSD (ours)</i>	0.653	0.081	0.606	0.099	0.649	0.138	0.646	0.111	0.669	0.147	0.645	0.171	0.633
	<i>VIG-TUQ (ours)</i>	0.664	0.067	0.620	0.091	0.657	0.136	0.656	0.107	0.674	0.145	0.645	0.070	0.652
SE		0.669	0.041	0.647	0.080	0.690	0.076	0.708	0.075	0.702	0.110	0.706	0.035	0.529
	<i>KLE</i>	0.682	0.055	0.654	0.078	0.700	0.074	0.716	0.069	0.715	0.110	0.701	0.030	0.643
VQARAD	<i>Log-Perplexity</i>	0.690	0.086	0.634	0.130	0.651	0.076	0.667	0.073	0.635	0.102	0.612	0.119	0.633
	<i>Token-Sar</i>	0.525	0.183	0.561	0.180	0.635	0.080	0.628	0.074	0.597	0.131	0.576	0.125	0.591
	<i>CCP</i>	0.665	0.120	0.613	0.155	0.638	0.096	0.655	0.099	0.639	0.062	0.615	0.084	0.604
	<i>Max-Pmb</i>	0.664	0.104	0.614	0.131	0.637	0.135	0.660	0.127	0.632	0.119	0.652	0.129	0.631
	<i>Token-Entropy</i>	0.606	0.086	0.638	0.126	0.662	0.077	0.673	0.089	0.639	0.098	0.618	0.122	0.632
	<i>VIG-TUQ-A (ours)</i>	0.722	0.084	0.680	0.102	0.674	0.074	0.686	0.054	0.664	0.096	0.640	0.113	0.667
	<i>VIG-TUQ-JSD (ours)</i>	0.721	0.085	0.682	0.113	0.646	0.082	0.679	0.054	0.659	0.109	0.603	0.117	0.660
	<i>VIG-TUQ (ours)</i>	0.727	0.055	0.685	0.106	0.672	0.073	0.691	0.057	0.668	0.109	0.640	0.113	0.669
SE		0.694	0.035	0.675	0.118	0.657	0.039	0.676	0.020	0.644	0.067	0.673	0.062	0.521
	<i>KLE</i>	0.716	0.036	0.687	0.118	0.689	0.042	0.699	0.028	0.672	0.066	0.699	0.065	0.616
VQARAD	<i>Log-Perplexity</i>	0.666	0.173	0.647	0.253	0.659	0.215	0.602	0.217	0.599	0.172	0.686	0.155	0.611
	<i>Token-Sar</i>	0.611	0.168	0.638	0.252	0.655	0.215	0.587	0.220	0.600	0.169	0.595	0.151	0.628
	<i>CCP</i>	0.661	0.141	0.653	0.230	0.608	0.196	0.593	0.186	0.588	0.113	0.657	0.136	0.589
	<i>Max-Pmb</i>	0.674	0.161	0.649	0.251	0.617	0.227	0.600	0.212	0.569	0.196	0.664	0.189	0.611
	<i>Token-Entropy</i>	0.609	0.169	0.636	0.256	0.657	0.223	0.605	0.211	0.605	0.166	0.605	0.123	0.617
	<i>VIG-TUQ-A (ours)</i>	0.713	0.164	0.677	0.254	0.672	0.215	0.708	0.211	0.607	0.174	0.706	0.152	0.634
	<i>VIG-TUQ-JSD (ours)</i>	0.616	0.185	0.540	0.270	0.689	0.213	0.670	0.200	0.665	0.140	0.636	0.148	0.631
	<i>VIG-TUQ (ours)</i>	0.713	0.164	0.677	0.254	0.693	0.213	0.670	0.200	0.659	0.141	0.706	0.152	0.641
SE		0.685	0.168	0.632	0.197	0.630	0.258	0.659	0.246	0.611	0.125	0.669	0.145	0.651
	<i>KLE</i>	0.679	0.169	0.630	0.198	0.644	0.259	0.678	0.247	0.620	0.126	0.666	0.144	0.679
ADVQA	<i>Log-Perplexity</i>	0.647	0.142	0.614	0.165	0.638	0.128	0.617	0.133	0.625	0.171	0.599	0.200	0.566
	<i>Token-Sar</i>	0.531	0.191	0.574	0.187	0.628	0.102	0.582	0.135	0.597	0.155	0.506	0.249	0.531
	<i>CCP</i>	0.643	0.109	0.603	0.119	0.649	0.088	0.637	0.100	0.633	0.134	0.063	0.250	0.588
	<i>Max-Pmb</i>	0.642	0.184	0.594	0.196	0.636	0.142	0.624	0.144	0.633	0.194	0.665	0.234	0.554
	<i>Token-Entropy</i>	0.642	0.145	0.612	0.173	0.639	0.124	0.614	0.138	0.624	0.175	0.594	0.246	0.556
	<i>VIG-TUQ-A (ours)</i>	0.665	0.129	0.627	0.162	0.651	0.114	0.624	0.133	0.644	0.166	0.617	0.233	0.577
	<i>VIG-TUQ-JSD (ours)</i>	0.620	0.146	0.617	0.156	0.642	0.109	0.613	0.138	0.594	0.182	0.515	0.232	0.531
	<i>VIG-TUQ (ours)</i>	0.663	0.129	0.633	0.155	0.662	0.117	0.633	0.130	0.642	0.164	0.617	0.233	0.567
SE		0.601	0.214	0.651	0.195	0.695	0.135	0.652	0.141	0.662	0.197	0.650	0.243	0.517
	<i>KLE</i>	0.614	0.213	0.660	0.194	0.701	0.132	0.651	0.139	0.666	0.196	0.662	0.242	0.568

Hyperparameter sensitivity and tuning requirements. As discussed above, VIG-TUQ requires only a lightweight hyperparameter search. To avoid relying on extensive tuning, we restrict the weighting coefficients α_{JSD} and α_A to integer values in $\{0, \dots, 5\}$, and tune them together with the attention layer l on a small training split. Table A.5 reports the selected hyperparameters when varying the number of training examples K . We observe that stable and competitive hyperparameters can already be obtained with as few as $K = 100$ examples. In several cases, such as Idefics-9B on OKVQA and VQARAD, the selected configuration remains identical across different values of K . Even when the selected coefficients vary, as for Qwen2.5-VL-3B on OKVQA, the resulting AUROC remains nearly unchanged. These results suggest that VIG-TUQ is not overly sensitive to the exact hyperparameter choice and does not require large validation sets or fine-grained tuning to achieve strong performance.

F Analysis of visual influence on hidden representations

In this section, we examine how the internal representations of an LVLMM change when visual input is removed. In particular, we compare the hidden representations obtained during normal generation with those obtained when the image is replaced by a blank input.

LVLMM. Let $\mathbf{x} = (\mathbf{x}_v, \mathbf{x}_l)$ be a multimodal input with a visual content $\mathbf{x}_v$ and a textual content $\mathbf{x}_l$. As described in Section 3, an LVLMM defines a parameterized function $f_\theta : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$ to encode the visual-linguistic context into a latent representation. In particular, the parametric function $f_\theta : \mathcal{X}_v \times \mathcal{X}_l \rightarrow \mathbb{R}^d$ can be defined recursively over L layers as

$$\begin{aligned}
 f_\theta(\mathbf{x}_v, \mathbf{x}_l) &= f_L(\mathbf{h}_{L-1}, \mathbf{x}_v), \quad \text{where} \\
 \mathbf{h}_i &= f_i(\mathbf{h}_{i-1}, \mathbf{x}_v) = g_i(m_i(E_i(\mathbf{x}_v), \mathbf{h}_{i-1})) \quad (1 \leq i \leq L) \\
 \mathbf{h}_0 &= E_l(\mathbf{x}_l);
 \end{aligned}$$

with $E_l(\cdot)$ a text embedding function. For each layer i , $g_i : \mathbb{R}^d \rightarrow \mathbb{R}^d$ depicts a transformer decoder layer, $m_i(\cdot, \cdot)$ denotes a multimodal fusion module (e.g., concatenation, linear layer, cross-attention layer), and $E_i(\cdot)$ is a visual encoder (e.g., ViT, CNNs, identity function).

Visual influence analysis problem. Our goal is to measure how much the hidden representations change at each layer l when the visual component $\mathbf{x}_v$ is removed. To this end, we examine the internal representations of the LVLMM that rely only on the textual component $\mathbf{x}_l$ defined as

$$\mathbf{h}'_i = f_i(\mathbf{h}_{i-1}, \emptyset) = g_i(m_i(\emptyset, \mathbf{h}'_{i-1})) \quad (1 \leq i \leq L) \quad (15)$$

$$\mathbf{h}'_0 = E_l(\mathbf{x}_l); \quad (16)$$

Table A.5: **Hyperparameter tuning stability under limited training data.** For each model–dataset pair, we tune the discrete weighting coefficients $\alpha_{\text{JSD}}, \alpha_A \in \{0, \dots, 5\}$ and the attention layer l using only K training examples. The selected hyperparameters and resulting AUROC/ECE remain stable even for small values of K , indicating that VIG-TUQ requires only lightweight tuning and is robust to small hyperparameter variations.

	Dataset	K	AUROC $\uparrow$	ECE $\downarrow$	α_{JSD}	α_A	l
idefics-9B	OKVQA	1000	0,645	0,159	0	1	16
	OKVQA	500	0,645	0,159	0	1	16
	OKVQA	200	0,645	0,159	0	1	16
	OKVQA	100	0,645	0,159	0	1	16
	VQARAD	1000	0,736	0,447	0	1	17
	VQARAD	500	0,730	0,375	1	4	16
	VQARAD	200	0,736	0,447	0	1	17
	VQARAD	100	0,736	0,447	0	1	17
Qwen2.5-VL-3B	OKVQA	1000	0,664	0,040	1	4	7
	OKVQA	500	0,664	0,014	3	4	7
	OKVQA	200	0,664	0,032	1	3	7
	OKVQA	100	0,664	0,014	3	4	7
	VQARAD	1000	0,713	0,127	0	1	6
	VQARAD	500	0,713	0,127	0	1	6
	VQARAD	200	0,713	0,127	0	1	6
	VQARAD	100	0,713	0,127	0	1	6

Table A.6

	Qwen2.5-VL-3B			Qwen2.5-VL-7B			llava-1.5-13B			llava-1.5-7B			idefics2-8B			idefics-9B			Qwen3.5-2B		
Dataset	α_{JSD}	α_A	l	α_{JSD}	α_A	l	α_{JSD}	α_A	l	α_{JSD}	α_A	l	α_{JSD}	α_A	l	α_{JSD}	α_A	l	α_{JSD}	α_A	l
OKVQA	1	4	7	0	1	4	2	3	6	2	3	16	1	4	29	0	1	16	0	1	5
VIZWIZ	3	1	9	2	3	15	1	4	16	3	4	8	1	1	20	0	1	14	1	3	5
VQARAD	0	1	6	0	1	6	3	1	3	1	0	18	3	2	0	0	1	17	1	1	5
ADVQA	1	4	7	1	3	4	1	4	6	1	3	4	1	4	1	0	1	14	1	3	0

To quantify the difference between these two embeddings at each layer i , we compute the cosine distance between $\mathbf{h}_i$ and $\mathbf{h}'_i$ at each layer. This distance reflects how much the model relies on visual information, where a larger value means that the image exerts a stronger influence on the internal representation. For each generated answer, we extract the hidden state corresponding to the last token at every layer. We then compute the cosine distance between the two representations and average the results across all samples of the dataset.

Analysis wrt prediction certainty. Figure A.6 shows the evolution of this distance across layers for certain and uncertain predictions. Several models are evaluated, including idefics2-8B, idefics-9B llava-1.5-13B, llava-1.5-7B, Qwen2.5-VL-3B, Qwen2.5-VL-7B and Qwen3.5-2B, all evaluated on **OKVQA** data-set. For all models, we observe that the distance between the two representations is larger for certain predictions than for uncertain ones. This means that when the model is confident about its answer, the internal representation changes more when the image is removed. In other words, confident predictions tend to rely more on visual information. Another observation is that the distance generally increases in deeper layers of the network. This suggests that the influence of visual information becomes stronger in later stages of the Transformer.

Analysis wrt prediction correctness. Figure A.7 compares the distance between representations for correct and incorrect predictions. For all models, we also observe that the distance between the two representations is larger for correct predictions than for incorrect ones. This indicates that when the model produces the right answer, its internal representation is more strongly affected by removing the image. In other words, correct predictions tend to rely more on visual information.

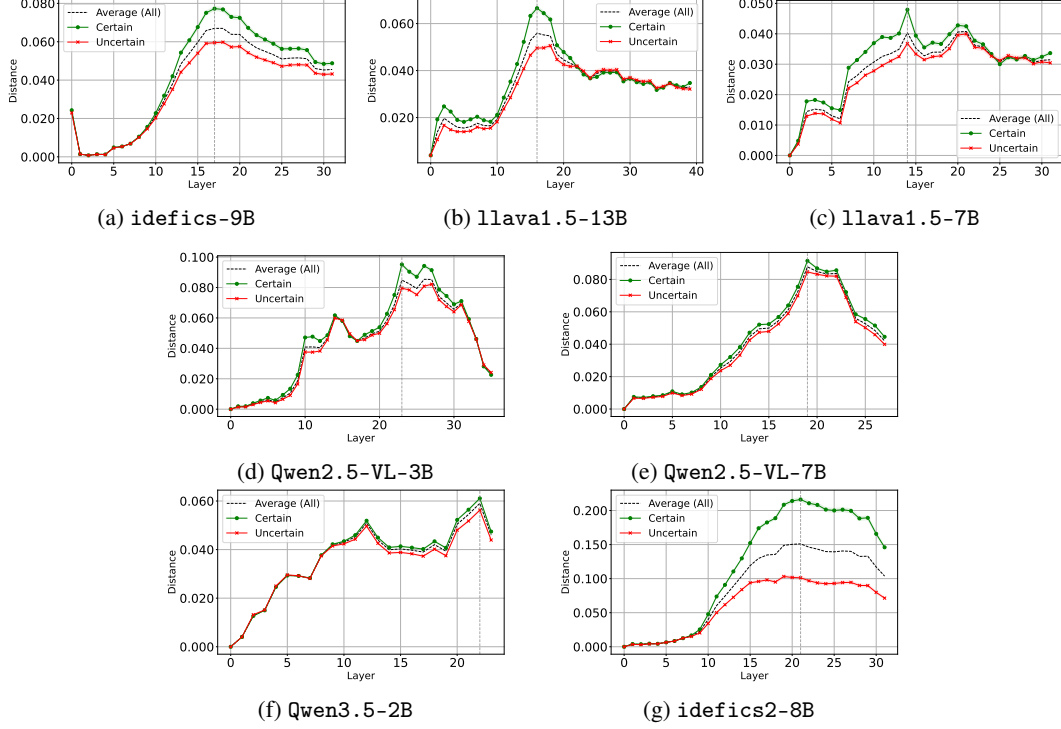


Figure A.6: **Layer-wise distance between hidden states from the original generation pass and the teacher-forced pass with a masked image.** The curves show the average similarity across layers for certain and uncertain samples for different LVLM architectures.

The separation between correct and incorrect predictions is generally slightly smaller than the one observed between certain and uncertain predictions, suggesting that visual reliance is more directly reflected in model confidence than in correctness. As in the certainty analysis, the distance also tends to increase in deeper layers of the network, indicating that visual information has a stronger influence in later Transformer layers.

Additional details on the spider plot. Figure 1 provides a compact aggregation of the layer-wise analyses reported in Figures A.6 and A.7. To summarize each comparison in the spider plot, we select the layer at which the absolute gap between the two group-wise mean curves is maximal. This selected layer corresponds to the dashed vertical line in Figures A.7 and A.6. We then extract the two group values at this layer, jointly normalize them within the corresponding comparison, and use them as the radial values for the associated model in Figure 1. Thus, each spoke in Figure 1 summarizes, for one model, the strongest layer-wise separation between the two groups, with larger polygon separation indicating that the image-induced representation shift is more discriminative of correctness or certainty.

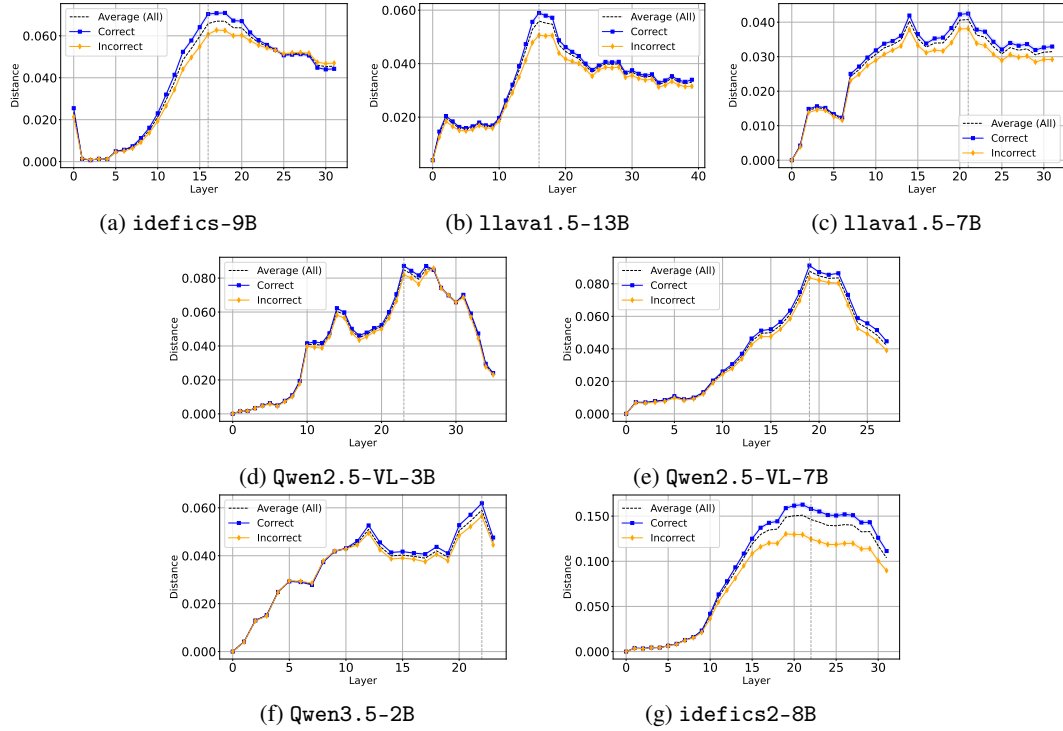


Figure A.7: **Layer-wise distance between hidden states from the original generation pass and the teacher-forced pass with a masked image.** The curves show the average similarity across layers for correct and incorrect samples for different LVLM architectures.